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UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING**



SUBJECT REPORT

Subject name: Web Development

(Subject code: CO3049)

SEMESTER 251 YEAR 2025-2026

- MAJOR: COMPUTER SCIENCE**
- PROGRAM: HIGH QUALITY**

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Ho Chi Minh City, 11/2025



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1 Introduction

The purpose of this assignment is to design and develop a personal blog web application. The main goal of this project is to create my own blog website where I can share my thoughts, life experiences, and lessons learned throughout my journey. Additionally, this project serves as an opportunity to apply the knowledge and techniques learned from the web development course.

The website was built from scratch using HTML, CSS, JavaScript, and PHP. It consists of multiple pages, including a homepage, an about page, a contact form, login/logout functionality, and a product listing page. The application integrates both front-end and simple back-end features to provide a complete and interactive web experience.

The development process focuses on applying front-end design principles, organizing content effectively, and ensuring usability across different devices through responsive design.

This report is organized as follows: Section 2 describes the system design and architecture. Section 3 presents the implementation details. Section 4 concludes the report with lessons learned during the development process and suggestions for future improvements.

2 Design Choices

2.1 Layout

I chose my layout like amazon website. The color choices follow very deliberate UX, branding, and psychological principles.

- Black used for text, navigation bars, headers and footers. It presents the strong, stable and neutral psychology. Moreover, black creates contrast with the white background, improving readability.
- Light background ensures product images remain the focus. Second, it makes the site feel bright, clean, and modern.
- Blue links ensures trust, reliability and calmness. Blue links are also a web-standard, improving intuitiveness.

To conclude, every color supports a purpose:

- High contrast improves accessibility.
- Blue links respect user expectations.
- White background maximizes clarity for product images.
- Green buttons give safety and positivity.

2.2 Diagrams

2.2.1 ERD (Entity-Relationship Diagram)

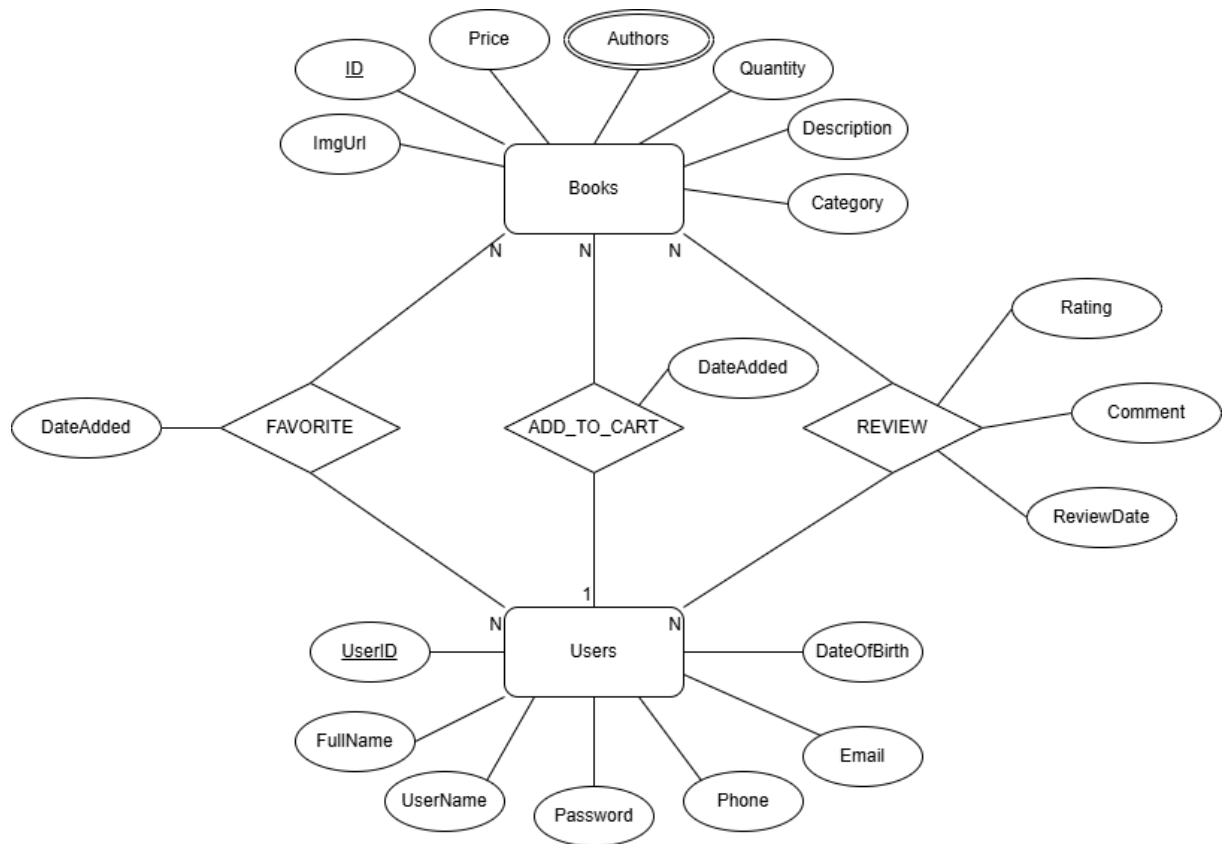


Figure 1: Caption

2.2.2 Sequence Diagram

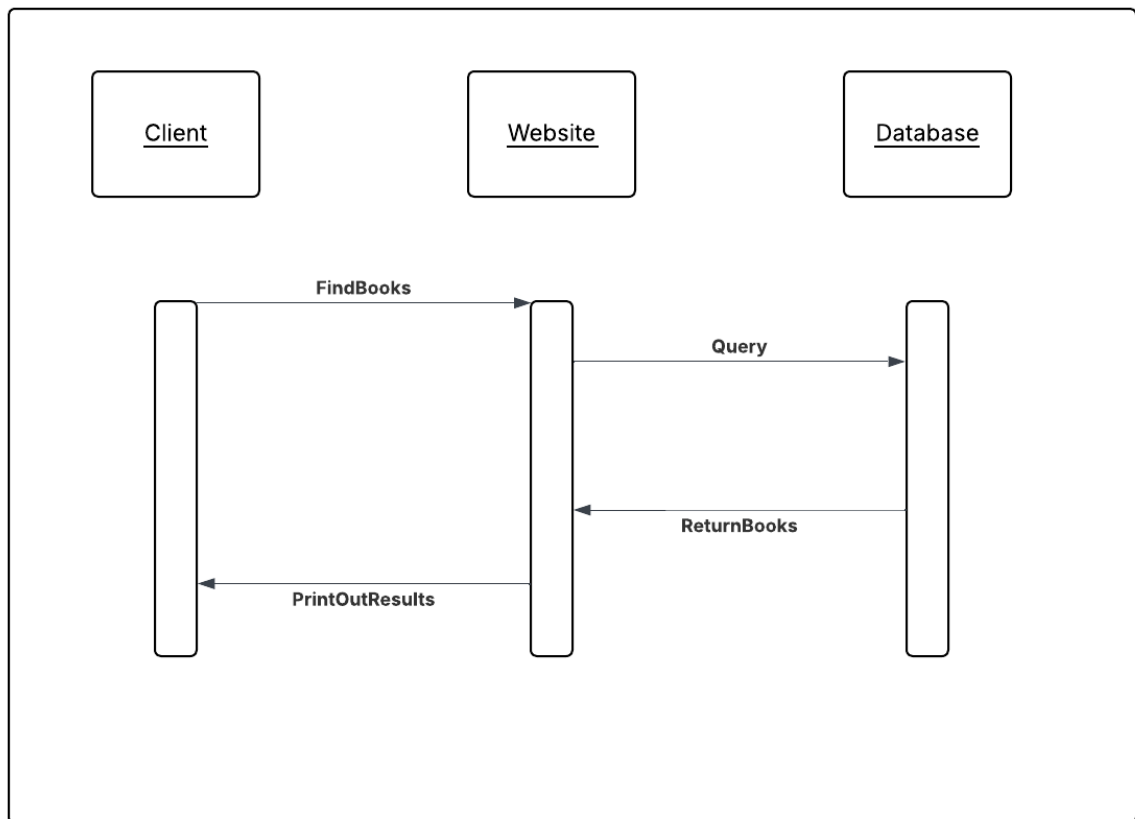


Figure 2: Caption

2.2.3 Use Case Diagram

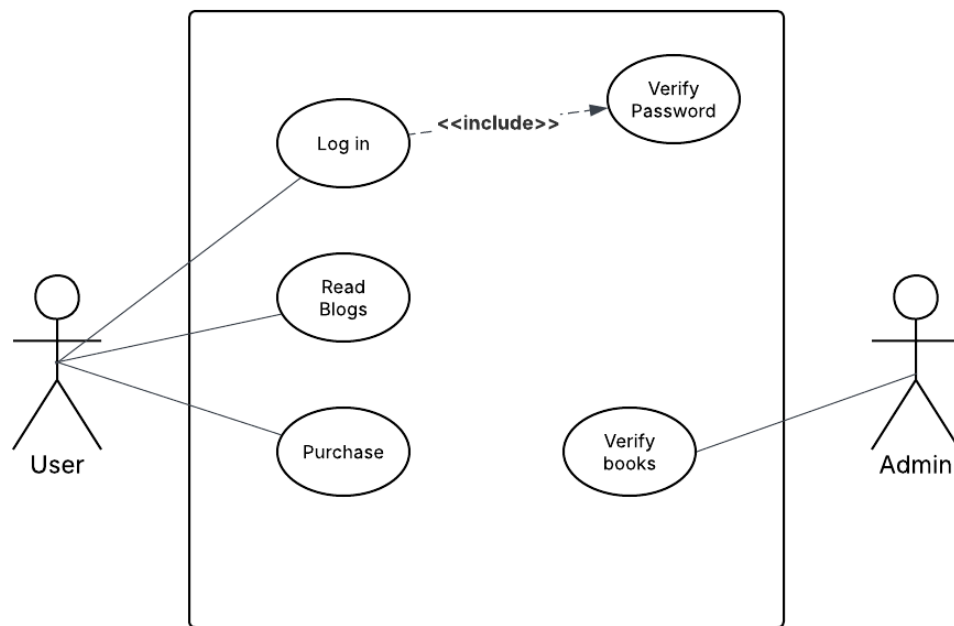


Figure 3: Caption

2.3 Web Pages

2.3.1 Header



Figure 4: Header of Website

2.3.2 Footer



Figure 5: Footer of Website

2.3.3 Home

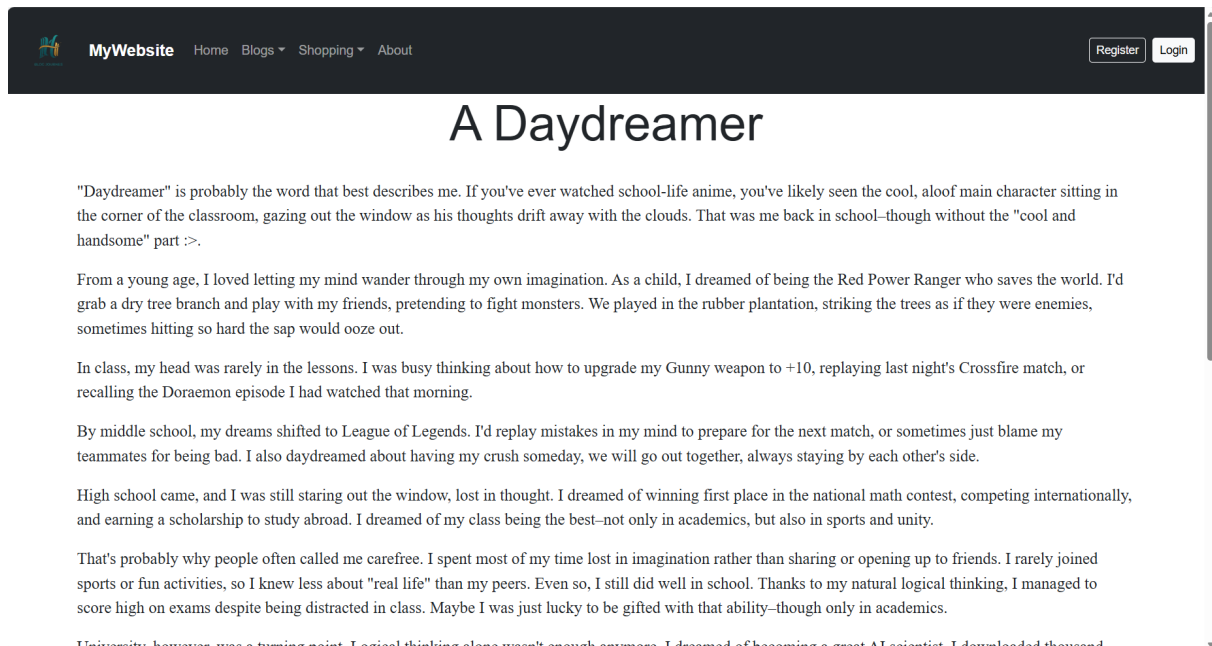


Figure 6: Home Page

2.3.4 Product Pages

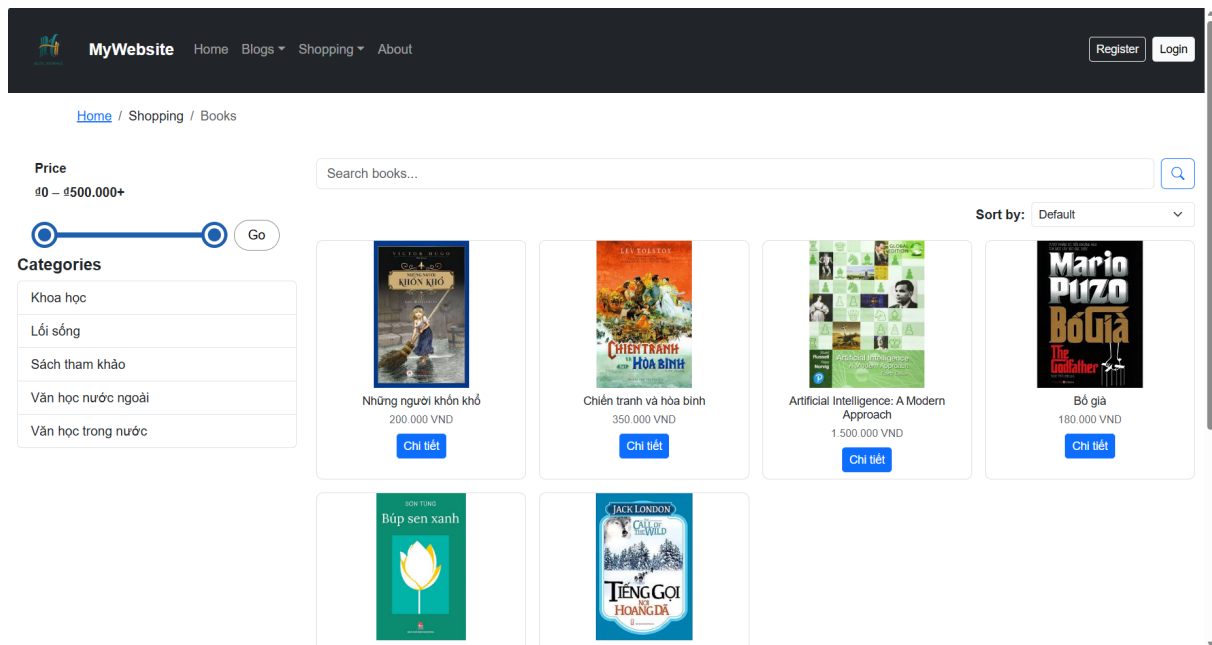


Figure 7: Product Page

2.3.5 Product Details

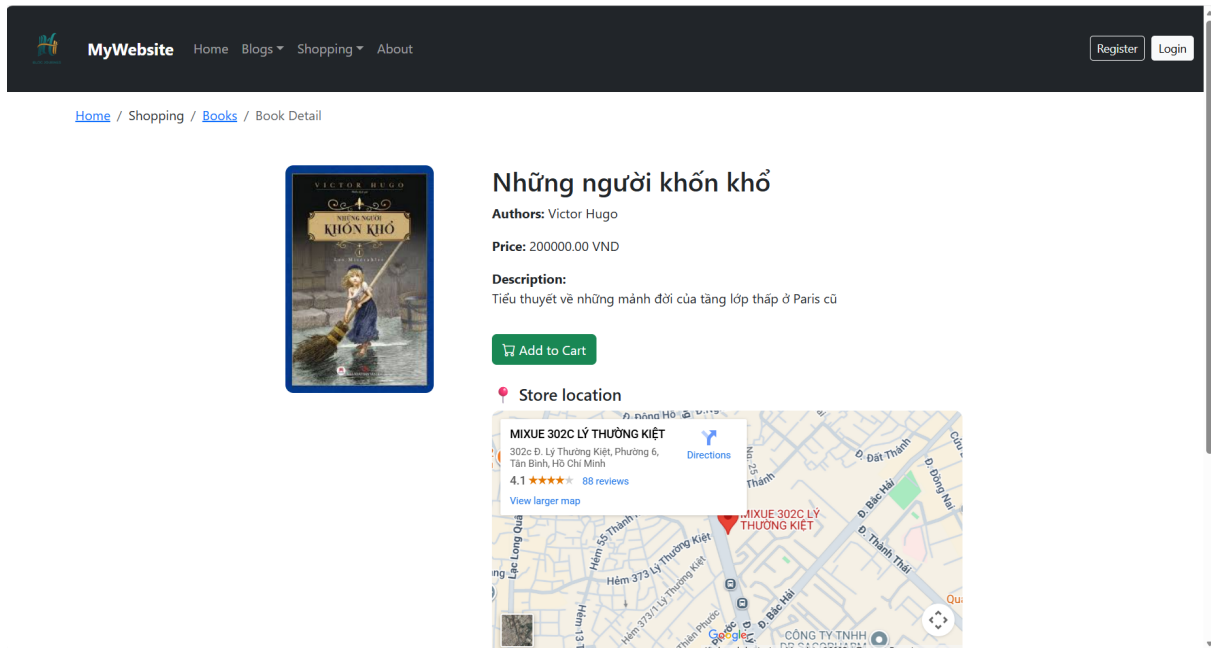


Figure 8: Product Details

2.3.6 About

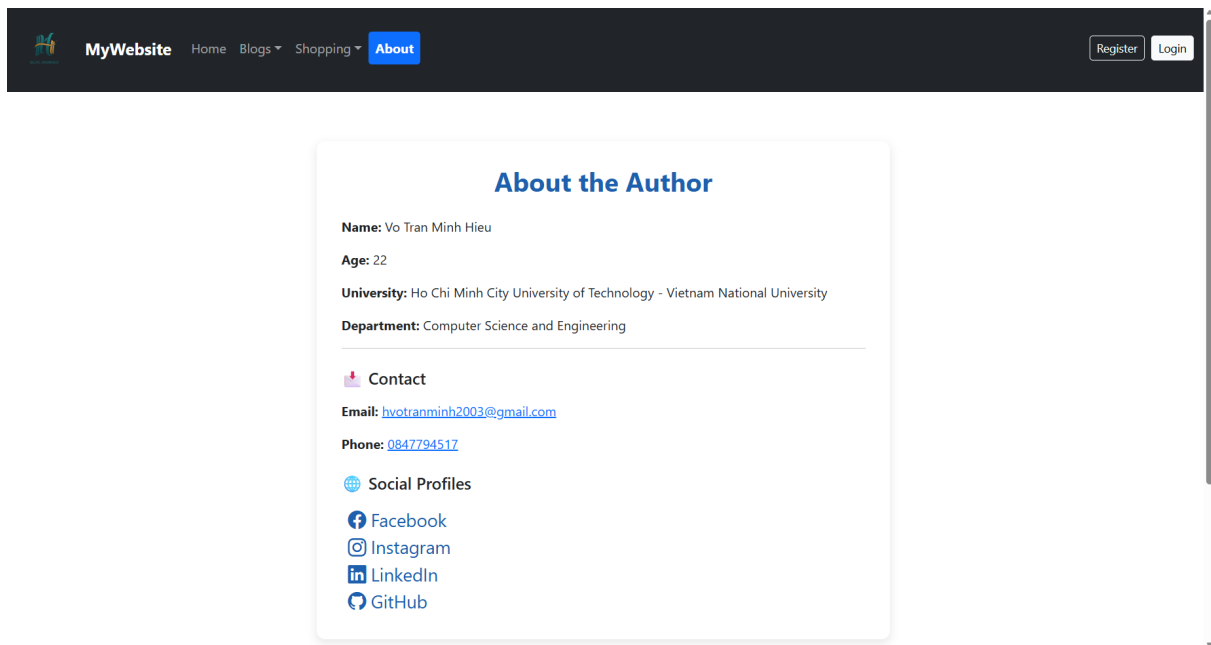
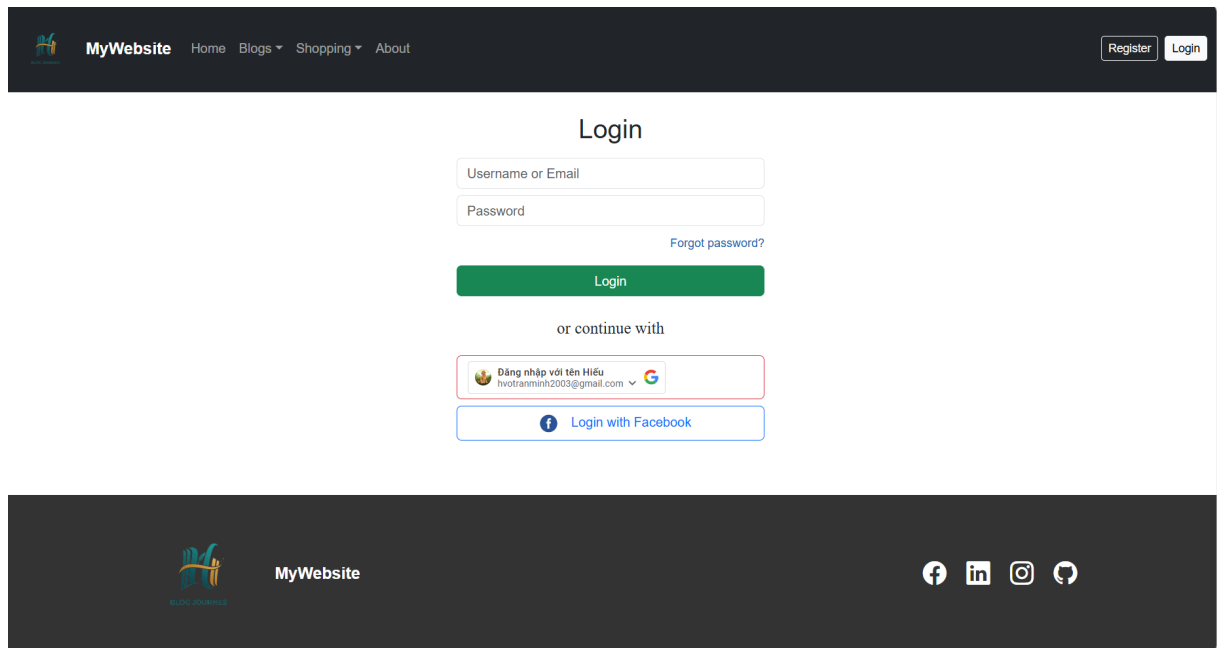


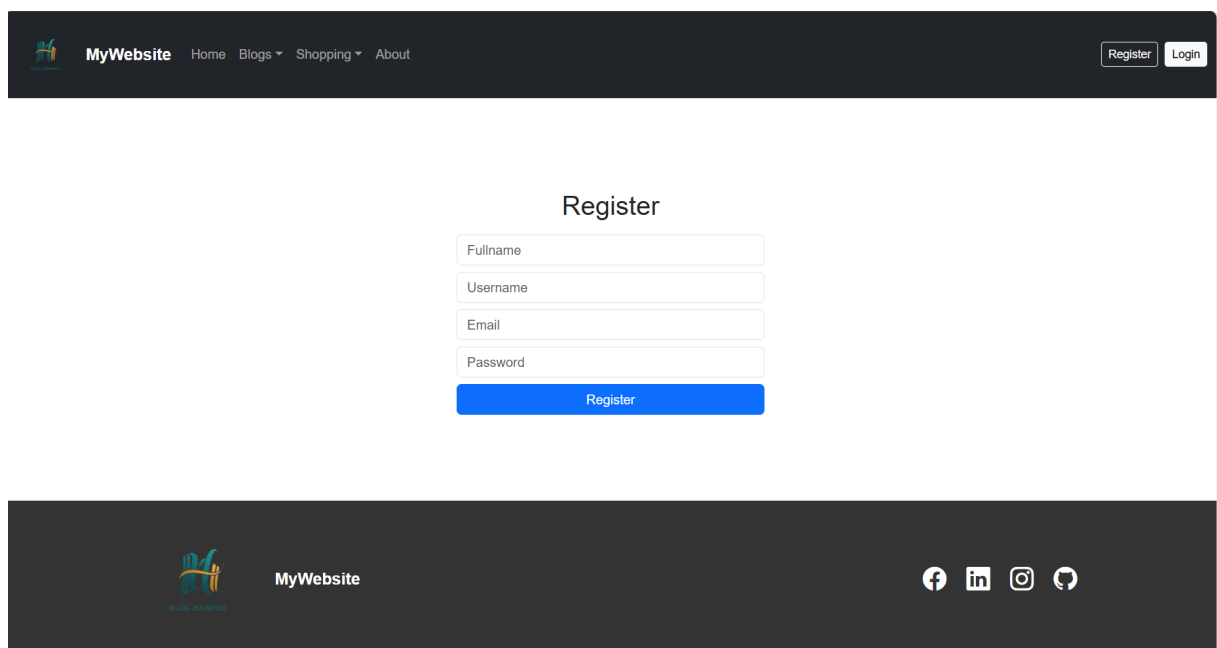
Figure 9: About

2.3.7 Authentication



The login page features a dark header with the 'MyWebsite' logo and navigation links (Home, Blogs, Shopping, About). On the right, there are 'Register' and 'Login' buttons. The main content area is titled 'Login' and contains two input fields for 'Username or Email' and 'Password'. A 'Forgot password?' link is positioned below the password field. A green 'Login' button is centered below the inputs. Below this, the text 'or continue with' is displayed. Two social login options are provided: a Google account link (showing 'Đăng nhập với tên Hiếu' and 'hvoctranminh2003@gmail.com') and a 'Login with Facebook' button. The footer includes the 'MyWebsite' logo, name, and social media icons for Facebook, LinkedIn, Instagram, and Twitter.

Figure 10: login



The register page has a similar dark header to the login page. The main content area is titled 'Register' and contains four input fields: 'Fullname', 'Username', 'Email', and 'Password'. A blue 'Register' button is located at the bottom of the form. The footer is identical to the login page, featuring the 'MyWebsite' logo, name, and social media icons.

Figure 11: register page

3 Implementation Details

3.1 Technology Stack

The application is built using a modern web development stack that emphasizes simplicity, performance, and maintainability.

3.1.1 Frontend Technologies

- **HTML5**: Semantic markup for improved accessibility and SEO
- **CSS3**: Custom styling with CSS variables for consistent theming
- **Bootstrap 5.3.2**: Responsive grid system and pre-built UI components
- **Bootstrap Icons**: Icon library for enhanced visual elements
- **Vanilla JavaScript (ES6+)**: Client-side interactivity and AJAX requests using modern Fetch API

3.1.2 Backend Technologies

- **PHP 7.4+**: Server-side scripting language for business logic and API endpoints
- **MySQL**: Relational database management system for data persistence
- **MySQLi**: PHP extension for database connectivity with prepared statements

3.1.3 Third-Party Libraries & Services

- **PHPMailer 7.0**: Email delivery system for user verification and password reset functionality
- **Google API Client 2.0**: OAuth integration for Google social login
- **Composer**: Dependency management for PHP packages

3.1.4 Development Environment

- **XAMPP**: Local development server (Apache + MySQL + PHP)
- **Git**: Version control system
- **GitHub**: Code repository hosting

3.2 Backend Implementation

The backend architecture follows a modular API-based approach, with separate PHP files handling specific functionalities.

3.2.1 Database Architecture

The application uses a MySQL database (blogsite) with the following key tables:

- **users:** Stores user account information including credentials and verification status
- **books:** Product catalog with book details, pricing, and images
- **categories:** Book categorization for filtering and navigation

Database Connection (db_connect.php):

```
1 $host = "localhost";
2 $user = "root";
3 $pass = "";
4 $dbname = "blogsite";
5 $conn = new mysqli($host, $user, $pass, $dbname);
```

3.2.2 User Authentication System

Registration Process

The registration system (register.php) implements comprehensive security measures:

3.2.2.1 Password Validation:

- Minimum 8 characters length
- At least one uppercase letter
- At least one numeric digit
- Server-side validation using regular expressions

```
1 if (strlen($password) < 8 ||
2     !preg_match('/[A-Z]/', $password) ||
3     !preg_match('/[0-9]/', $password)) {
4     echo json_encode([
5         "success" => false,
6         "message" => "Password must be at least 8 characters, include a
           number and an uppercase letter."
7     ]);
8     exit;
9 }
```

3.2.2.2 Security Features:

- **Password Hashing:** Uses PASSWORD_BCRYPT algorithm for secure password storage
- **Email Verification:** Generates unique verification tokens to confirm email ownership
- **Duplicate Prevention:** Checks for existing email addresses before registration
- **Prepared Statements:** Prevents SQL injection attacks

3.2.2.3 Email Verification Flow:

1. User submits registration form
2. System generates random 32-character verification token
3. PHPMailer sends verification email with unique link
4. User clicks link to activate account
5. Token is validated and account status updated

Login System

The login system (login.php) features:

3.2.2.4 Authentication Process:

- Accepts username or email for login flexibility
- Verifies credentials using password_verify() function
- Creates secure PHP sessions upon successful authentication
- Returns JSON responses for AJAX integration

3.2.2.5 Legacy Password Migration: The system includes automatic migration from legacy password formats (MD5, SHA1, plaintext) to modern bcrypt hashing:

```
1 if ($password_verify_ok && $migrated && is_string($stored_hash)) {  
2     $new_hash = password_hash($password, PASSWORD_DEFAULT);  
3     if ($new_hash !== false) {  
4         $up = $conn->prepare("UPDATE users SET password_hash=? WHERE userid  
5                               =?");  
6         $up->bind_param("si", $new_hash, $user['userid']);  
7         $up->execute();  
8     }  
}
```

3.2.2.6 Error Handling:

- Custom error handlers convert PHP errors to JSON responses
- Output buffering prevents accidental whitespace breaking JSON format
- Detailed error messages for debugging (can be disabled in production)

Password Recovery

The forgot password system (forgot_password.php) implements:

- Token-based password reset links
- Email delivery via PHPMailer with SMTP configuration
- SSL certificate verification handling for local development
- Time-limited reset tokens for security

3.2.3 AJAX Search Implementation

The search functionality (search_books.php) provides real-time book search capabilities:

3.2.3.1 Backend API:

```
1 $q = trim($_GET['q'] ?? '');
2 if ($q === '') {
3     echo json_encode([]);
4     exit;
5 }
6 $stmt = $conn->prepare("SELECT id, title, image_url, price FROM books WHERE
    title LIKE CONCAT('%', ?, '%') LIMIT 8");
7 $stmt->bind_param("s", $q);
8 $stmt->execute();
9 $result = $stmt->get_result();
10 $suggestions = [];
11 while ($row = $result->fetch_assoc()) {
12     $suggestions[] = $row;
13 }
14 echo json_encode($suggestions);
```

3.2.3.2 Key Features:

- **Prepared Statements:** Prevents SQL injection in search queries
- **LIKE Operator:** Enables partial matching for flexible search
- **Result Limiting:** Returns maximum 8 suggestions for performance
- **JSON Response:** Structured data for frontend consumption

3.2.3.3 Frontend Integration (product_book.html):

```
1 searchInputLive.addEventListener('input', async () => {
2     const query = searchInputLive.value.trim();
3
4     if (query.length < 2) {
5         dropdown.style.display = 'none';
6         return;
7     }
8 }
```

```
9   try {
10     const res = await fetch('api/search_books.php?q=${encodeURIComponent(
11       query)}');
12     const data = await res.json();
13     dropdown.innerHTML = data.map(item => '
14       <a href="book_detail.html?id=${item.id}"
15         class="list-group-item list-group-item-action search-result-item">
16         ${item.title}
17       </a>
18     ').join('');
19     dropdown.style.display = 'block';
20   } catch (err) {
21     console.error('Search error:', err);
22   }
23 });
```

3.2.3.4 Search Features:

- **Live Search:** Real-time suggestions as user types (minimum 2 characters)
- **Debouncing:** Prevents excessive API calls during typing
- **Autocomplete Dropdown:** Displays search results in a dropdown menu
- **Click-to-Navigate:** Direct links to book detail pages
- **Outside Click Detection:** Closes dropdown when clicking elsewhere

3.2.4 Additional Backend Features

Session Management (check_session.php):

- Validates active user sessions
- Returns user information for frontend display
- Enables persistent login state across pages

Book Catalog API (books.php):

- Retrieves books with price range filtering
- Category-based filtering
- Pagination support for large datasets

Category Management (categories.php):

- Dynamic category listing
- Used for sidebar navigation filters

3.3 Frontend Implementation

The frontend emphasizes user experience, responsive design, and modern web standards.

3.3.1 Design Philosophy

The application follows a clean, professional design approach with emphasis on:

- **Readability:** Clear typography and adequate spacing
- **Consistency:** Unified color scheme and component styling
- **Responsiveness:** Mobile-first design using Bootstrap grid system
- **Accessibility:** Semantic HTML and ARIA labels

3.3.2 Color Scheme & Theming

Primary Color Palette (style.css):

```
1 :root {  
2   —primary: #1e5fae;           /* Professional blue */  
3   —track-bg: #e6e6e6;          /* Light gray for UI elements */  
4   —thumb-size: 24px;           /* Consistent sizing */  
5   —track-height: 6px;          /* Slider track height */  
6 }
```

3.3.2.1 Color Psychology:

- **Primary Blue (#1e5fae):** Conveys trust, professionalism, and reliability
- **Neutral Grays:** Provides visual hierarchy without overwhelming users
- **White Backgrounds:** Ensures readability and clean appearance

3.3.2.2 Application:

- Navigation elements and CTAs use primary blue
- Hover states provide visual feedback
- Active states highlight current page/selection
- Consistent button styling across all forms

3.3.3 Typography

3.3.3.1 Font Selection:

- **Primary Font:** Arial, sans-serif (system font for fast loading)
- **Body Text:** Times New Roman for blog content (classic, readable)
- **Headings:** Large, bold Arial for clear hierarchy

3.3.3.2 Font Sizing:

```
1 h1 {  
2   font-size: 60px;  
3   text-align: center;  
4   margin-bottom: 32px;  
5 }  
6 p {  
7   font-size: 20px;  
8   font-family: "Times New Roman", Times, serif;  
9   margin-bottom: 18px;  
10 }
```

3.3.4 Layout & Components

Responsive Navigation

Features:

- Bootstrap navbar with collapse functionality for mobile
- Dynamic active state highlighting based on current page
- Session-aware display (Login/Register vs. User Profile/Logout)
- Dropdown menus for categorized content

Dynamic Loading:

```
1 fetch("navbar.html?nocache=" + new Date().getTime())  
2   .then(response => response.text())  
3   .then(html => {  
4     document.getElementById("navbar-placeholder").innerHTML = html;  
5  
6     const script = document.createElement("script");  
7     script.src = "js/sessionCheck.js";  
8     script.onload = () => updateNavbar('index.html');  
9     document.body.appendChild(script);  
10  });
```

Product Cards

Book Card Design (product_book.html):

```
1 .book-card {  
2   border: 1px solid #ddd;  
3   border-radius: 8px;  
4   transition: transform 0.2s, box-shadow 0.2s;  
5 }  
6 .book-card:hover {  
7   transform: translateY(-4px);  
8   box-shadow: 0 3px 8px rgba(0, 0, 0, 0.1);  
9 }
```

Features:

- Hover animations for interactivity
- Consistent image sizing with object-fit
- Price formatting with Vietnamese locale
- Direct links to detailed product pages

Custom Range Slider

Price Filter Implementation:

- Dual-thumb range slider for min/max price selection
- Custom styling with CSS variables
- Real-time label updates showing selected range
- “Go” button to apply filters

Visual Design:

```
1 .slider input[type="range"]::-webkit-slider-thumb {  
2   width: var(--thumb-size);  
3   height: var(--thumb-size);  
4   border-radius: 50%;  
5   background: var(--primary);  
6   border: 4px solid #fff;  
7   box-shadow: 0 0 0 4px var(--primary);  
8   cursor: pointer;  
9 }
```

3.3.5 Interactive Features

Client-Side Filtering:

- Category selection with visual active states
- Price range filtering with live updates
- Search-based filtering with pagination
- Sorting options (price, title, ascending/descending)

Pagination:

- Smart pagination with ellipsis for large datasets

- Previous/Next navigation
- Direct page number selection
- Responsive to filter changes

Form Validation:

- Real-time password strength validation
- Email format validation
- Required field indicators
- User-friendly error messages

3.3.6 User Experience Enhancements

Loading Optimization:

- Cache-busting for dynamic content (?nocache= timestamp)
- Lazy loading of navbar and footer components
- Minimal external dependencies

Accessibility:

- Semantic HTML5 elements (<nav>, <main>, <aside>)
- ARIA labels for screen readers
- Keyboard navigation support
- Sufficient color contrast ratios

Mobile Responsiveness:

- Bootstrap grid system (col-6, col-md-4, col-lg-3)
- Flexible layouts that adapt to screen size
- Touch-friendly button sizes
- Collapsible navigation for small screens

3.4 SEO Implementation

The application implements the following SEO techniques to improve search engine visibility and social media sharing:

3.4.1 On-Page SEO

- Unique, descriptive title tags for each page
- Meta descriptions (150-160 characters)
- Meta keywords targeting relevant search terms
- Robots meta tags (index, follow)
- Canonical URLs to prevent duplicate content

3.4.2 Social Media SEO

- Open Graph Protocol meta tags (Facebook, LinkedIn)
- Twitter Card meta tags
- Custom social sharing images and descriptions

3.4.3 Structured Data (Schema.org)

- Organization schema markup
- Contact information schema
- Social media profile linking (sameAs property)

3.4.4 Technical SEO

- XML sitemap for search engine crawling
- Robots.txt file for crawler directives
- SEO-friendly URL structure
- Breadcrumb navigation
- Performance optimization (CDN, minimal dependencies, optimized images)

3.4.5 Content SEO

- Semantic HTML5 elements
- Proper heading hierarchy (H1 → H2 → H3)
- Descriptive alt text for images
- Meaningful anchor text for links

3.4.6 Mobile SEO

- Responsive viewport meta tag
- Mobile-first design with Bootstrap
- Touch-friendly interface elements

Summary

The MyWebsite application demonstrates a comprehensive implementation of modern web development practices:

- **Backend:** Secure, API-driven architecture with robust authentication and real-time search
- **Frontend:** Responsive, user-friendly interface with modern design principles
- **SEO:** Multi-layered optimization strategy for search engines and social media
- **Security:** Password hashing, prepared statements, email verification, and session management

The complete implementation can be accessed in the GitHub repository: [Github link](#).

4 Conclusion

4.1 Lessons Learned

Throughout the course and the process of completing this assignment, I have gained substantial knowledge and practical experience across various aspects of software development and project management.

I learned to use **GitHub** for version control and collaborative development, which enhanced my understanding of source code management, branching strategies, and issue tracking. This provided me with essential skills for maintaining structured and organized software projects.

From a technical perspective, I developed proficiency in **HTML**, **CSS**, and **JavaScript**, which enabled me to design and implement interactive and visually appealing web interfaces. On the backend, I acquired foundational skills in **PHP**, which allowed me to build server-side logic and manage communication between the **frontend**, **backend**, and **database** layers.

Furthermore, I learned to design and implement **relational databases**, write **SQL queries**, and perform data management tasks efficiently. I also utilized the **Bootstrap** framework to streamline the process of creating responsive web layouts. Through this, I gained an appreciation for the importance of appropriate layout design, including the selection of suitable dimensions, colors, and structures to ensure a consistent user experience.

In developing the **product page**, I implemented **pagination** and **lazy loading** to enhance performance and optimize data retrieval. Additionally, I integrated **AJAX (Asynchronous**

JavaScript and XML) to enable seamless communication between the client and server. This approach improved responsiveness by allowing the webpage to update dynamically without requiring a full reload, thereby providing a smoother user experience.

Beyond core development, I explored the integration of **external APIs** such as **Google Maps** and **Facebook**, expanding the functionality and interoperability of the application. I also implemented **user authentication features**, including user registration, login, and password recovery through email verification, to ensure secure access control.

Finally, I studied **Search Engine Optimization (SEO)** techniques to improve the visibility and accessibility of the website to potential users.

In summary, this assignment provided me with comprehensive, hands-on experience in **end-to-end web development**—from user interface design and backend logic to database management and deployment. It also allowed me to document and describe the system through **database schemas** and **software architecture diagrams**, reinforcing both my technical and analytical skills in software engineering.

4.2 Future Improvements

In the future, I plan to implement the following improvements:

- Add detailed content for the remaining incomplete pages.
- Implement a shopping cart and payment methods.
- Develop a user information page that includes basic profile data, account balance, and starred (favorite) products.
- Implement features that allow me to create new articles and users to create new reviews.
- Complete the properties outlined in the ERD.